

POPULATION MIGRATION ANALYSIS TO DKI JAKARTA

Understanding Migrant Origins, Destination Patterns, and Migration Routes

Data Analyst Case Study

Data Period: 2025

1. Executive Summary

Population migration is an important aspect of understanding population distribution and movement within an urban area. DKI Jakarta attracts population movement from various provinces and cities, particularly from surrounding areas on Java Island.

This project analyzes population migration into DKI Jakarta during 2025 using publicly available population migration data. The analysis focuses on three main dimensions: where migrants come from, where they settle within DKI Jakarta, and which origin-destination routes have the highest migrant volumes.

The project applies an end-to-end data analytics workflow consisting of data cleaning, exploratory data analysis, SQL analysis, and interactive visualization using Power BI.

The dataset records a total of 110,327 migrants. The analysis shows that West Java is the largest origin province in the dataset, while Kota Bekasi is the largest origin city. Within DKI Jakarta, Kota Administrasi Jakarta Timur receives the largest number of migrants. The highest individual migration route identified is Kota Bekasi → Kota Administrasi Jakarta Timur, with 3,585 migrants.

The resulting dashboard provides an interactive overview of migration patterns and can be used as a basis for understanding population movement between surrounding areas and DKI Jakarta.

2. Background

DKI Jakarta is a major metropolitan center with strong economic, employment, educational, and social connections to surrounding regions.

Population movement into Jakarta does not occur uniformly. Migrants originate from different provinces and cities and settle across different administrative areas within DKI Jakarta.

Understanding these patterns can provide useful context for population distribution and urban planning analysis.

However, simply knowing the total number of migrants does not provide enough insight. It is also important to understand:

- Where migrants come from
- Which provinces contribute the most migrants
- Which cities contribute the most migrants
- Which areas of Jakarta receive the most migrants
- Which districts and villages have higher concentrations
- Which origin-destination routes have the highest migration volumes

This project was developed to answer these questions through a structured data analytics workflow.

3. Problem Statement

The main problem addressed in this analysis is:

How are migrants entering DKI Jakarta distributed by their origin, destination, and migration route during 2025?

The analysis is designed to identify migration concentration patterns and major origin-destination flows.

4. Analytical Objectives

The objectives of this project are:

1. Measure the total number of migrants recorded in the dataset.
2. Identify the provinces from which migrants originate.
3. Identify the cities that contribute the largest number of migrants.
4. Analyze the distribution of migrants across DKI Jakarta's administrative cities.
5. Identify districts with the highest migrant concentrations.
6. Identify villages with the highest migrant concentrations.
7. Identify the most common migration routes between origin and destination areas.
8. Present the findings through an interactive Power BI dashboard.

5. Key Business Questions

The analysis focuses on the following questions:

1. Overall Migration
 - How many migrants are recorded in the dataset?
2. Origin Analysis
 - Which provinces contribute the most migrants?
 - Which origin cities contribute the most migrants?
3. Destination Analysis
 - Which administrative cities in DKI Jakarta receive the most migrants?
 - Which districts receive the most migrants?
 - Which villages receive the most migrants?
4. Migration Route Analysis
 - Which origin-destination routes have the highest migrant volumes?
 - Which origin areas have strong migration flows toward specific areas of Jakarta?

6. Dataset

- Dataset Name = Data Penduduk Pendetang ke Provinsi DKI Jakarta
- Data Period = 2025
- Data Source = Satu Data Jakarta
- Dataset Description

The dataset contains information about population migration into DKI Jakarta, including origin and destination geographical information and the number of migrants associated with each origin-destination combination.

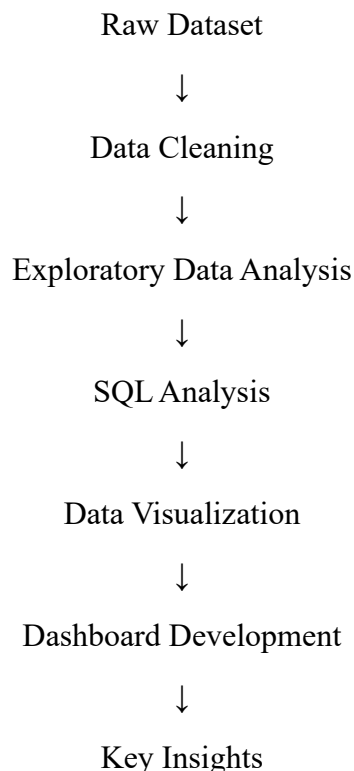
The analysis uses geographical attributes such as:

- Origin Province
- Origin City
- Destination City
- Destination District
- Destination Village
- Total Migrants

A derived field called Migration Route is also used to represent the movement from an origin city to a destination city.

The dataset is analyzed at an aggregated origin-destination level, where each record represents the number of migrants associated with a specific combination of origin and destination areas.

The project follows an end-to-end analytics process:



The tools used throughout the project are:

Tool	Purpose
Python	Data cleaning and exploratory analysis
Pandas	Data manipulation
Google Colab	Data analysis workflow
SQL	Analytical queries and aggregation
Power BI	Interactive dashboard and visualization
GitHub	Project documentation and version control

8. Data Cleaning

Before performing the analysis, the dataset was reviewed and prepared for analysis.

The main data preparation steps included:

5. 8.1 Data Inspection

The dataset structure, columns, data types, and records were reviewed to understand the available information.

6. 8.2 Data Type Validation

Numerical and categorical fields were checked to ensure that they could be used correctly during analysis.

The `total_migrants` field was treated as a numerical variable so that aggregation and statistical analysis could be performed correctly.

7. 8.3 Missing Value Checking

The dataset was checked for missing or incomplete values in relevant fields.

8. 8.4 Categorical Data Standardization

Categorical geographical fields were reviewed to ensure consistent values and naming conventions.

Relevant fields include:

- `origin_province`
- `origin_city`
- `destination_city`
- `destination_district`

- destination_village

9. 8.5 Migration Route Creation

A derived field called **Migration Route** was created by combining the origin city and destination city.

Example:

Kota Bekasi → Kota Adm. Jakarta Timur

This field was used to analyze the most common migration flows.

10. 8.6 Data Validation

The cleaned dataset was reviewed to ensure that the resulting values could be aggregated correctly and used consistently across Python, SQL, and Power BI.

9. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the overall distribution of migration records before building the final dashboard.

The analysis focused on:

- Total migrants
- Origin provinces
- Origin cities
- Destination cities
- Destination districts
- Destination villages
- Migration routes

10. Overall Migration Volume

The dataset records: 110,327 migrants

This represents the total number of migrants contained in the analyzed dataset for 2025.

The total migrant count is used as the primary KPI in the Power BI dashboard.

11. Origin Province Analysis

The analysis shows that West Java is the largest origin province in the dataset.

West Java contributes approximately 37K migrants or approximately 33.84% of the total migrant population recorded in the dataset.

Other major origin provinces include:

- Central Java
- Banten
- East Java
- North Sumatra
- Lampung

This distribution indicates that migration into DKI Jakarta is strongly connected to surrounding regions, particularly provinces located on Java Island.

12. Origin City Analysis

At the origin-city level, Kota Bekasi records the largest number of migrants among the cities shown in the analysis.

The leading origin cities include:

1. Kota Bekasi
2. Kota Depok
3. Bogor
4. Kota Tangerang
5. Bekasi
6. Kota Tangerang Selatan

Kota Bekasi records approximately 6.6K migrants, while Kota Depok and Bogor each account for approximately 5K migrants.

The concentration of major origin cities around the Jakarta metropolitan area indicates the importance of migration flows between Jakarta and its surrounding urban areas.

13. Destination Analysis

Among the destination cities, Kota Administrasi Jakarta Timur records the largest migrant count.

Destination City	Migrants
Kota Adm. Jakarta Timur	32K
Kota Adm. Jakarta Selatan	25K
Kota Adm. Jakarta Barat	25K
Kota Adm. Jakarta Utara	18K
Kota Adm. Jakarta Pusat	10K

14. Destination District Analysis

At the district level, the dashboard identifies several areas with relatively high migrant counts.

The leading destination districts include:

- Cakung
- Cengkareng
- Kalideres
- Jagakarsa
- Cilincing

Among these, Cakung records the highest migrant count in the displayed Top Destination Districts analysis. This provides a more detailed view than the administrative-city analysis by showing where migrants are concentrated at a smaller geographical level.

15. Destination Village Analysis

The analysis is further broken down into destination villages.

The leading destination villages displayed in the dashboard include:

- Kapuk
- Pulo Gebang
- Penjaringan
- Kalideres
- Penggilingan

Among the displayed villages, Kapuk records the highest migrant count. This level of analysis provides a more granular perspective on population concentration within DKI Jakarta.

16. Migration Route Analysis

The largest migration route identified in the dataset is Kota Bekasi → Kota Administrasi Jakarta Timur with: 3,585 migrants

Migration Route	Migrants
Kota Bekasi → Kota Adm. Jakarta Timur	3,585
Kota Depok → Kota Adm. Jakarta Selatan	2,226
Kota Depok → Kota Adm. Jakarta Timur	1,780
Kota Tangerang Selatan → Kota Adm. Jakarta Selatan	1,712
Kota Tangerang → Kota Adm. Jakarta Barat	1,672

17. Origin-Destination Relationship

The analysis also examines the relationship between origin provinces and destination cities.

A matrix visualization was created in Power BI to show migrant volumes across:

Origin Province × Destination City

This allows the analysis to identify how migration from particular provinces is distributed across different areas of DKI Jakarta. For example, West Java contributes a large number of migrants across multiple destination cities, while other provinces have comparatively smaller migration volumes. This origin-destination view provides additional context that cannot be obtained by looking at origin or destination totals independently.

18. Power BI Dashboard

The final Power BI dashboard consists of three analytical pages.

Page 1 — Population Migration Analysis

The first page provides an executive overview of the dataset.

It contains:

- Total Migrants
- Number of Origin Provinces
- Number of Origin Cities
- Number of Destination Cities
- Number of Destination Districts
- Number of Destination Villages
- Top Origin Provinces
- Top Origin Cities
- Migrants by Destination City

Page 2 — Destination Analysis

The second page focuses on where migrants settle within DKI Jakarta.

It contains:

- Top Destination Districts
- Top Destination Villages
- Migrants by Destination City
- Origin Province × Destination City matrix

Page 3 — Migration Route Analysis

The third page focuses on migration flows between origin and destination areas.

It contains:

- Top Migration Routes
- Migration Route Details

19. Key Findings

Based on the analysis, several key findings were identified.

1. Finding 1 — Migration volume
The dataset contains a total of 110,327 migrants for the 2025 observation period.
2. Finding 2 — West Java is the largest origin province
West Java contributes approximately 37K migrants, representing around 33.84% of the total migrants recorded in the dataset. This makes West Java the largest origin province in the analyzed data.
3. Finding 3 — Bekasi is a major origin city
Kota Bekasi records approximately 6.6K migrants, making it the largest origin city shown in the analysis. Kota Depok and Bogor also represent significant origin locations.
4. Finding 4 — Jakarta Timur receives the largest migrant volume
Kota Administrasi Jakarta Timur records approximately 32K migrants, the highest destination-city total shown in the dashboard. Jakarta Selatan and Jakarta Barat follow with approximately 25K migrants each.
5. Finding 5 — Migration is concentrated in several districts
At the district level, Cakung records the highest migrant count among the displayed destination districts. Other highly represented districts include Cengkareng, Kalideres, Jagakarsa, and Cilincing.
6. Finding 6 — Several villages show higher concentrations
Kapuk records the highest migrant count among the displayed destination villages, followed by Pulo Gebang, Penjaringan, Kalideres, and Penggilingan.
7. Finding 7 — The largest migration route connects Bekasi and Jakarta Timur
The largest individual migration route identified is Kota Bekasi → Kota Administrasi Jakarta Timur with 3,585 migrants. This route represents one of the most prominent migration flows in the dataset.

20. Recommendations

The findings can be used to support further population and urban planning analysis.

Focus further analysis on high-concentration destination areas

Areas with relatively high migrant concentrations can be prioritized for deeper analysis of population growth, housing, transportation, public facilities, and other urban services.

The current analysis identifies destination cities, districts, and villages with relatively high migrant volumes, but additional datasets would be required to determine specific infrastructure needs.

Monitor major migration corridors

Major migration routes such as:

Kota Bekasi → Jakarta Timur

can be examined further to understand recurring population movement between surrounding cities and DKI Jakarta.

Additional transportation and commuting datasets could provide further context.

Combine migration data with other datasets

Migration data alone provides information about population movement but does not explain all underlying causes.

Future analysis could combine this dataset with:

- Population density
- Housing availability
- Employment
- Transportation
- Income
- Education
- Public facilities
- Land use

This would allow more comprehensive analysis of the factors associated with migration patterns.

Develop time-series analysis in future datasets

The current dataset covers 2025.

If migration data from multiple years becomes available, future analysis could examine:

- Year-over-year migration changes
- Emerging origin areas
- Changes in destination preferences
- Changes in migration routes
- Long-term population movement patterns

This would make it possible to identify trends rather than only describing the 2025 distribution.

21. Limitations

Several limitations should be considered when interpreting this analysis.

- **Single-year dataset**
The analysis covers 2025 only. Therefore, the project describes the migration distribution for the available period rather than long-term migration trends.
- **Aggregated data**
The dataset is aggregated by geographical origin and destination rather than representing individual migrant-level records. Therefore, the analysis cannot directly examine individual demographic characteristics.
- **Limited explanatory variables**
The dataset primarily provides geographical migration information and migrant counts. It does not by itself explain why people migrate or which socioeconomic factors influence migration.
- **Migration does not necessarily represent permanent settlement**
The available migration records should be interpreted according to the definition and methodology of the original dataset. The analysis should not automatically assume that every recorded migrant represents permanent relocation.

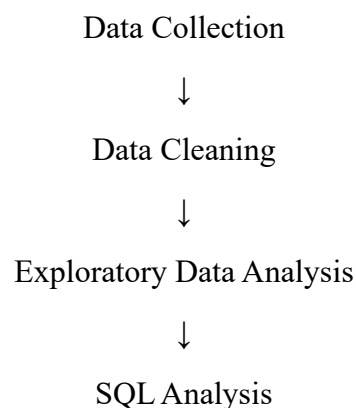
22. Conclusion

This project analyzes population migration into DKI Jakarta during 2025 using an end-to-end data analytics workflow. The analysis identifies the major origins, destination areas, and migration routes within the available dataset.

A total of 110,327 migrants are recorded. West Java is the largest origin province, while Kota Bekasi is the largest origin city shown in the analysis. Within DKI Jakarta, Kota Administrasi Jakarta Timur receives the largest migrant volume. At a more granular level, several districts and villages show higher concentrations of migrants.

The analysis also identifies Kota Bekasi → Kota Administrasi Jakarta Timur as the largest migration route, with 3,585 migrants. The Power BI dashboard transforms these findings into an interactive analytical tool consisting of an executive overview, destination analysis, and migration route analysis.

Overall, this project demonstrates an end-to-end data analyst workflow:





Data Visualization



Power BI Dashboard



Insights & Recommendations

The analysis can be further developed by incorporating multi-year migration data and socioeconomic, transportation, housing, and population datasets to provide a broader understanding of population mobility into DKI Jakarta.